EXPERIMENT 13: Word Frequency Distribution Using Python

Aim

To write a Python program to read a text document named sample_text.txt, calculate the frequency of each word, and display the word frequency distribution.

Algorithm

1. Open the text file sample_text.txt.
2. Read the text and convert it to lowercase.
3. Remove punctuation and split the text into words.
4. Count the frequency of each word using a dictionary.
5. Display the words and their frequencies.

Program

Writing

```
import string

file = open("sample_text.txt", "r")
text = file.read().lower()
file.close()

for ch in string.punctuation:
    text = text.replace(ch, "")

words = text.split()
frequency = {}

for word in words:
    frequency[word] = frequency.get(word, 0) + 1

print("Word Frequency Distribution:")
```

```
for word, count in frequency.items():
    print(word, ":", count)
```

Sample Text File

Create a text file named **sample_text.txt** containing:

Python is easy. Python is powerful.

Data analysis is easy.

Important: Save sample_text.txt in the same folder as your Python program.

Sample Output

Word Frequency Distribution:

python : 2

is : 3

easy : 2

powerful : 1

data : 1

analysis : 1

output

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 13.py =====
Word Frequency Distribution:
python : 2
is : 3
easy : 2
powerful : 1
data : 1
analysis : 1
```

Result

The Python program successfully reads the text document and displays the frequency distribution of its words.